

Tumours, risk, and waves: learning from multi-modal data to predict outcomes in *NF2* schwannomatosis

Chay Paterson

University of Manchester

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Where I am and what I do

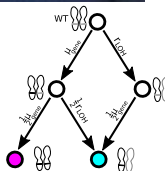
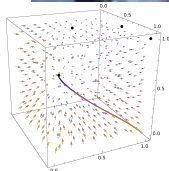
Smith Lab
(Manchester Royal
Infirmary) →



My modelling group
(University of Manchester)



← Wedge Lab and
Jamie Weaver's
group (The
Christie/MCRC)



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Schwannomatosis and radiotherapy

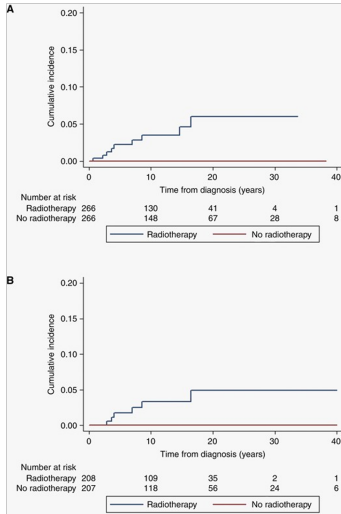
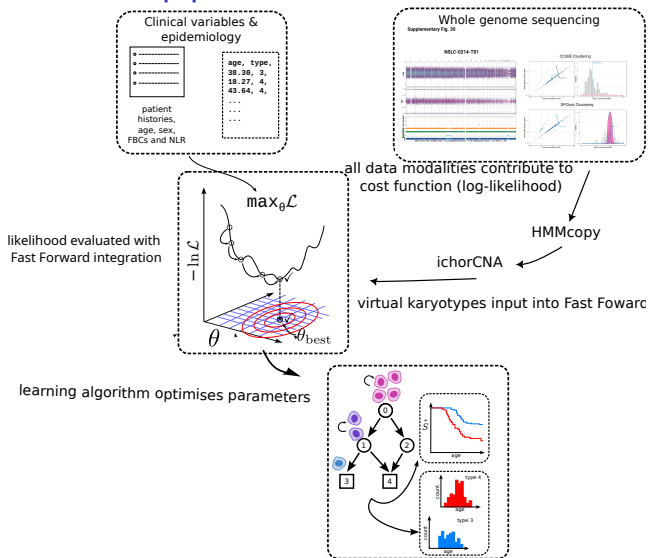


Figure 2. Kaplan-Meier for central nervous system malignancy (9 radiotherapies, 0 no radiotherapy) using matched cases for matched cases (n = 266 radiotherapy) and controls (n = 266 no radiotherapy) Log-rank Chi-square = 10.02, P = .0016.

- Complication: secondary CNS/skull base malignancies
- Excess absolute risk of 5%
- Relative risk more than 10× general pop
- (Conditional) mean recurrence time 6y

“NF2 patients should not be offered radiotherapy as first-line treatment of benign tumors and should be given a frank discussion of the potential 5% excess absolute risk of M/MP.”¹

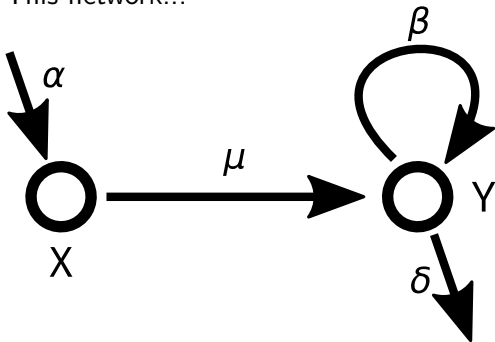
Idealised pipeline for multimodal data integration



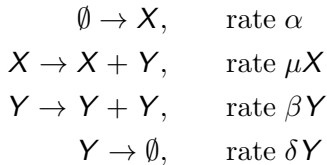
The problem is that existing models are not general enough to include both modalities (in a nontrivial likelihood function)

A brief history of multi-stage clonal expansion theory

This network...

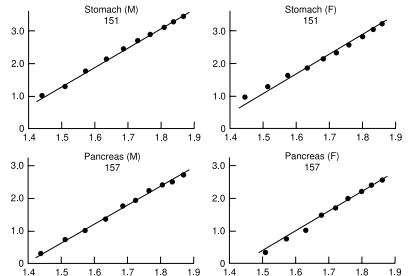


corresponds to this
stochastic process:...



A brief history of multi-stage clonal expansion theory

P. Armitage and R. Doll¹²



¹P. Armitage and R. Doll, British Journal of Cancer 1954; 8: 1–12

²P. Armitage and R. Doll, British Journal of Cancer 1957; 11(2): 161-169

A brief history of multi-stage clonal expansion theory

A.G. Knudson¹²

Mutation and Retinoblastoma 823

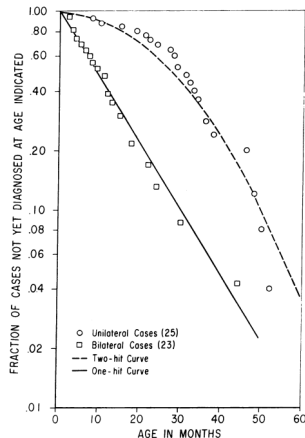
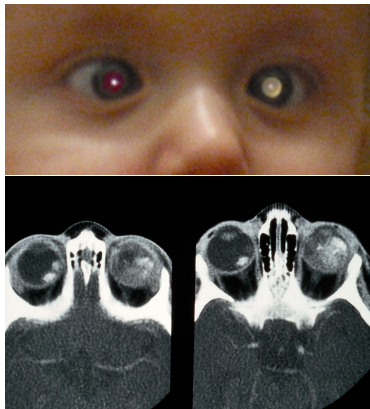


FIG. 1. Semilogarithmic plot of fraction of cases of retinoblastoma not yet diagnosed (S) vs. age in months (t). The one-hit curve was calculated from $\log S = -t/30$, the two-hit curve from $\log S = -4 \times 10^{-6} t^2$.

¹AG. Knudson, PNAS 68.4 (1971): 820-823.

²F. Michor, Y. Iwasa and MA. Nowak, Nature Reviews Cancer 2004; 4: 197-205 doi:10.1038/nrc1295

How do these studies work?

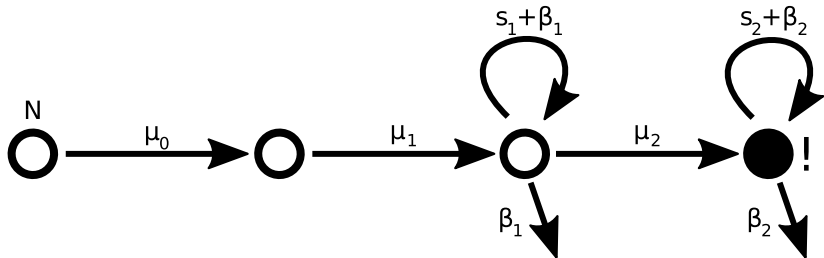
What are the relevant observables in this type of longitudinal study & data analysis?

Track a cohort that initially contains N patients in the study:

- ▶ Age-specific incidence $I(a)$: rate at which new cases are recorded in the cohort with ages between a and $a + da$
- ▶ Diagnosis-free survival function $S(a)$: probability to survive to age a without being diagnosed.

$$I(a) = -N \frac{dS}{da} = -N S(a) \frac{d \ln S}{da} \quad (1)$$

So what?

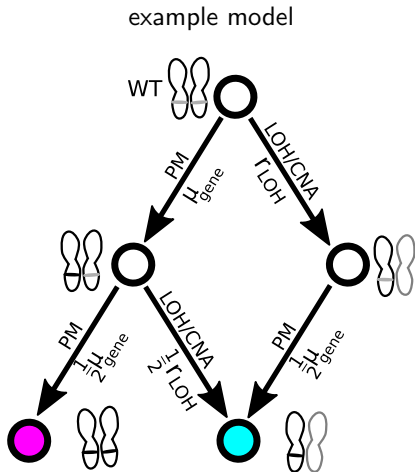


- ▶ The survival curve $S(a)$ determines the incidence curve $I(a)$
- ▶ The model determines the survival curve: the probability not to end up at one of the end nodes of the graph. We want to know what model+parameters best agree with data from studies.
- ▶ To fit (or “train”) the model to longitudinal data, we need to compute $S(a)$!

This is the central mathematical problem in cancer epidemiology.
How do we compute S ?

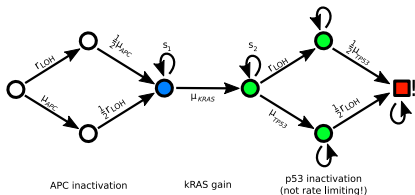
Models on graphs

1. Most methods for computing $S(a)$ do not consider graphs with multiple end nodes
2. To study **specific genes** and mechanisms of interest (SNVs, LOH, CNA, etc.), we need to evaluate $S(a)$ for a model defined on a graph (right)



This gets us the incidence of *specific genomic subtypes*

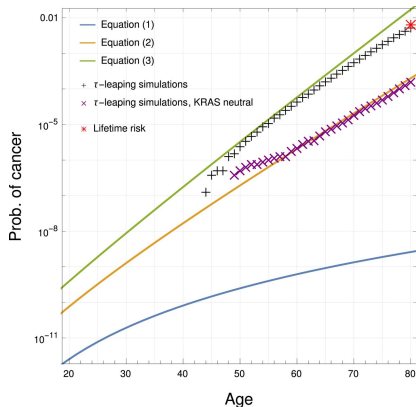
Colorectal adenocarcinoma model



- ▶ Can use A-D type approximation, or stochastic simulations
- ▶ Conditional path probabilities $P(X_i)$ encode fitnesses

but:

- ▶ Mean-field breaks down at old ages / large probabilities
- ▶ Stochastic simulations are *extremely slow*



Alternative approach: Kolmogorov forward equations

Define a more general generating function Ψ :

$$\Psi(t, \vec{q}) = \mathbb{E} \left[\prod_j q_j^{N_j} \right] \quad (2)$$

and derive Kolmogorov forward equations instead. Then we can numerically integrate these, and get survival curves $S_i(a)$ for different types of cancer i . E.G. tumours with clonal LOH, or no clonal LOH.

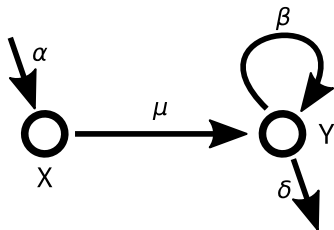
$$S_i(a) = \frac{\Psi(a, \vec{q}_F)}{\Psi(a, \vec{q}_{F-i})} \quad (3)$$

with F end nodes. People knew about this approach for a long time (since 1970s) but it was never considered as useful as backward equations.

¹SH Moolgavkar and DJ Venzon, Math. biosc. 1979; 47(1): 55-77

²DW Quinn, Risk Analysis 1989; 9(3): 407-13

Kolmogorov forward equations as wave equations



Briefly: the Kolmogorov forward equations in $P(N_0, \dots)$

$$\begin{aligned} \frac{dP(N_0, N_1, \dots)}{dt} = & \sum_{\text{vertices}} \alpha(N_j - 1)P(\dots, N_j - 1, \dots) \\ & - \alpha N_j P(\dots, N_j, \dots) \\ & + \beta \dots + \mu \dots + \delta \dots \end{aligned}$$

get transformed...

Kolmogorov forward equations as wave equations

They transform with $P \rightarrow \Psi$:

$$\frac{\partial \Psi}{\partial t} = \sum_{\text{vertices}} \alpha(q_j - 1) q_j \frac{\partial \Psi}{\partial q_j} + \dots = \mathcal{H} \Psi \quad (4)$$

where $\mathcal{H}$ is a hyperbolic differential operator.

Because this is a hyperbolic wave equation, we can solve for future values of Ψ if we have initial values, by evolving them along characteristics.

¹SH Moolgavkar and DJ Venzon, Math. biosc. 1979; 47(1): 55-77

Kolmogorov forward equations

Using the big generating function Ψ , find the corresponding wave equation:

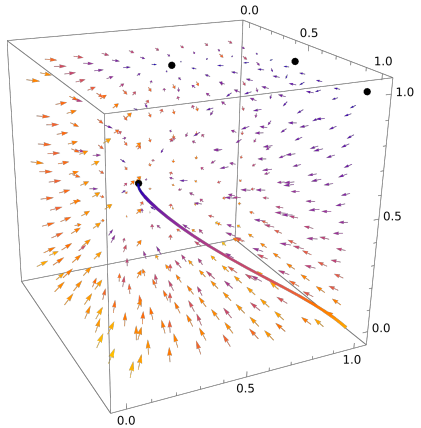
$$\frac{d\Psi}{dt} = \mathcal{H}\Psi \quad (5)$$

This can be solved using the method of characteristics, so we can solve

$$\frac{d\vec{\gamma}}{dt} = \vec{X} \quad (6)$$

numerically, using an appropriate time stepper.

(NB: (6) is always a Hamiltonian system too)



The vector field $\vec{X}$ and a characteristic $\vec{\gamma}$

Errors and runtime

To estimate $\hat{S}$ to within an error less than chosen ϵ , we would need to run $T \propto \epsilon^{-0.5}$ Monte Carlo simulations.

What happens with our new method, using a simple Runge-Kutta stepper?

$$\epsilon \sim \Delta t^2 \tag{7}$$

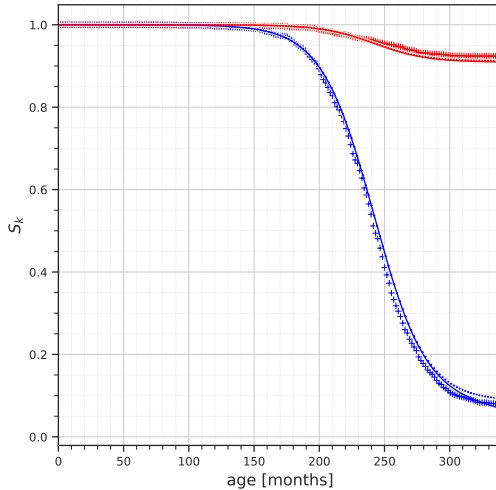
and runs in

$$T \sim \Delta t^{-1} \sim \mathcal{O}(\epsilon^{-1/2}) \tag{8}$$

so new runtime $\sim \mathcal{O}(\text{old runtime}^{1/4})$.

Amazing!

Fast forward method

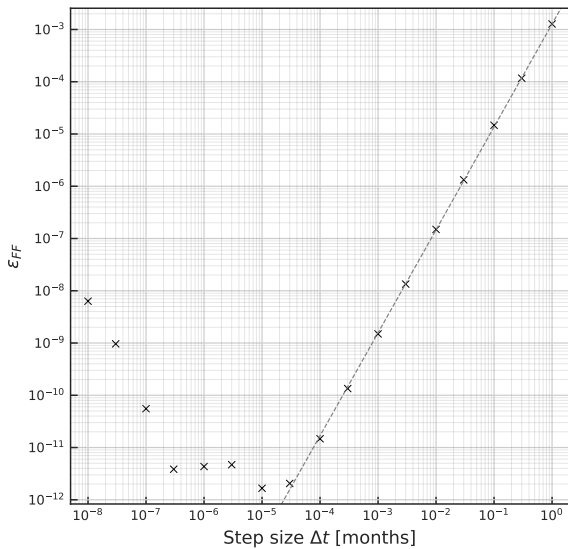


Random sampling vs fast forward method:

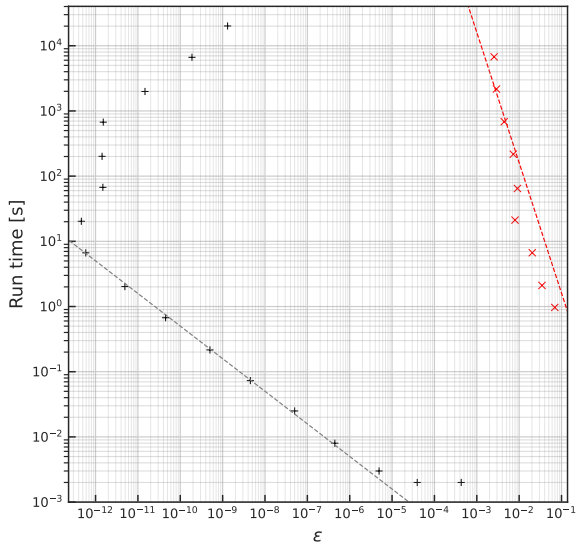
Monte Carlo: $\approx 5000s$ Fast forward: 4ms

Fast forward method

Error analysis

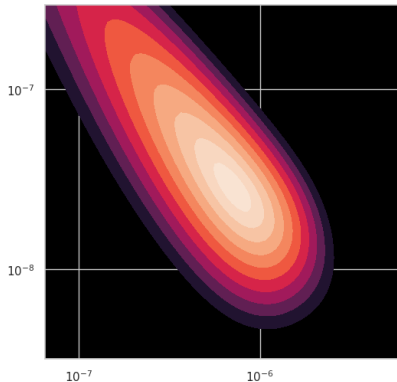
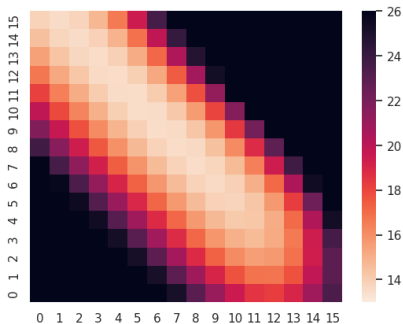


How do they compare?



Fast forward method

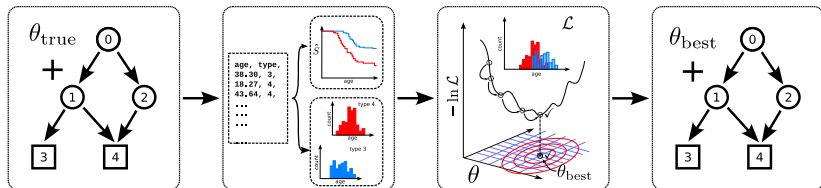
Efficiently computing $S_k(a)$ means we can evaluate likelihood functions directly, just sampling them.....



Machine learning with the fast forward method

The training harness

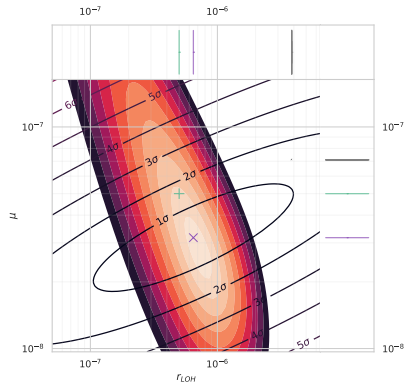
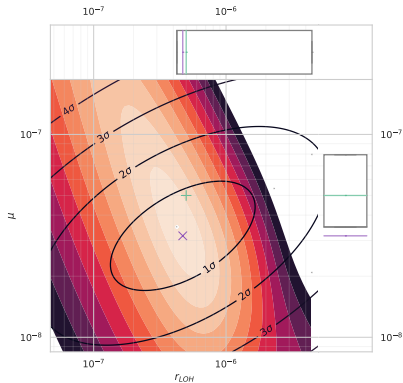
Here, “machine learning” means we minimise a function: in this case the negative log-likelihood function $-\ln \mathcal{L}$



generate a simulated study, then find the max. likelihood parameter estimates

Machine learning with the fast forward method

Parameter inference



Initial findings from the schwannomatosis project

- ▶ Fitness dis/advantages of single hits to *NF2* tumour suppressor practically unidentifiable
- ▶ Tumour commitment in *NF2*-schwannomatosis occurs very early
- ▶ *NF2* loss appears to be *sufficient* for a primary tumour, with other alterations occurring after

Conclusions and the future

- ▶ Immediate clinical applications *without pharmaceutical products*, just better advice about interventions. Both the oncologist and patient benefit: clarity about uncertainty/risks is good for trust and planning.
- ▶ There are deep theoretical connections too: Hamiltonian systems are v common, I suspect because of irreversibility.
- ▶ It now appears (as of 2026) that many auto-immune diseases may also be caused by clonal expansions too¹

¹Pantelis Nicola et al., Nature 654, 131-41 (2026)

Acknowledgements



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- ▶ Ivana Božić (here!)
- ▶ Gareth Evans
- ▶ David Wedge



The University of Manchester